

Duck Hero

The official GDD for my INDEPRO2 GameBoy student project game, by Matéo Costille

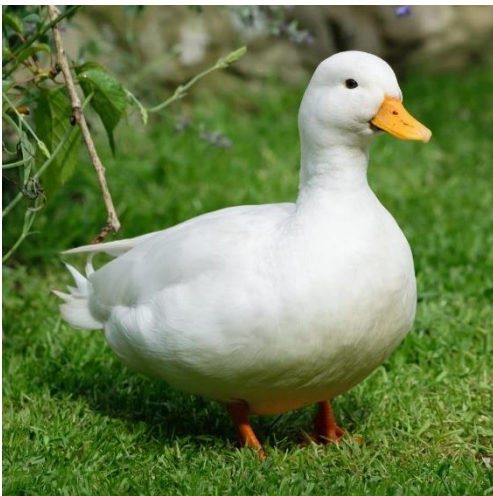


Concept of the game:

The player controls a Duck Knight from a top-down view with 3 health points. The duck has 4 different weapons available to him, to which he can swap between whenever he needs. He has to use the different weapons he has to save his kingdom.

Game lore:

The king inquiries about our player character, the Duck knight, and asks him to save the Duck Kingdom from the Lynx Emperor and his Lynx Army. To do so the player will need to embark on an Epic adventure through 4 levels which will take him from the Duck Castle to the very heart of Lynx territory: the lynx Fortress.



Mechanics of the game:

Weapons

The knight will have four different weapons:

- Sword
- Bow
- Shield
- Hammer

Here are the different properties for the different weapons:

Sword:

The sword is the most basic weapon but also the most versatile one: it allows the player to kill anything that is in front of him. It has infinite use and has the fastest attack speed, but suffers from a very short range.

Bow:

The bow allows the player to shoot an arrow in the desired direction but doing so consumes an arrow. He can carry a maximum of 9 arrows on him at once, and the only way to gain arrows is by cutting grass, which has a 1/3 chance of spawning an arrow that the player may pick up. The bow's attack speed is slightly slower than the sword's attack speed, and this coupled with the low arrow count means that the bow isn't very good at dealing with multiple enemies.

Shield:

The shield protects the player from all damage coming from enemy arrows, and when equipped it allows the player to push rocks around (the player braces his shield and slams with all of his weight into the rock, shield first. This pushes the rock until it collides with something).

Hammer:

The hammer is the only weapon that offers crowd control to the player: he can slam it to the ground to make a shock which roots enemies for a full second. They will still be able to damage you if they come into contact with you, but they will remain immobile for the full second.

	Attack Speed	Range	Specialty	Max ammo
Sword	6.6/s	3 block	*	*
Bow	2.8/s	Full screen	Consume ammo	9
Shield	*	*	Can push rocks	*
Hammer	0.5/s	4 block AOE	Roots enemies	*

* means no information

Actors

There are different actors other than the player which can be placed into the level. Each of these actors has different uses. Here is a list of all the actors:

- Enemy Lynx
- Boss
- Grass
- Rocks
- Locks
- Keys
- HealthPoints
- HUD
- ArrowHUD

First of all, we will clear the first misconception out: The three last actors cited, HealthPoints, HUD and ArrowHUD, are “Pinned” actors, which means they have no collision and are constantly pinned on a specific part of the screen. This is useful to create Dynamic UI, but as they aren’t actors which interact with the player character we will bypass them for now and talk about them in another part of the GDD.

Enemy Lynx:

We can’t have a good Hero without good enemies. The Lynx are all part of the Lynx Emperor’s army, and all actively search to defeat the player. There are 3 types of enemy Lynx: the basic type, which follows the player, the horizontal archer which walks from top to bottom and shoots right or left toward the player, and the vertical archer which walks from right to left and shoots up or down toward the player.

Boss:

The boss is the actor which represents the Lynx Emperor. He is seen only once on the final level (boss level). He is faster than the other Lynx but is part of the “basic” category since he simply runs toward the player.

Grass:

These actors aren’t simply used as decoration but is very useful for the player as it is the one and only way to get arrows. When grass is cut, there is a 1/3 chance that it will drop an Ammo pick-up which restores one arrow.

Rocks:

Seemingly insignificant, these actors start off as simple obstacles which block the player and force them to use their shield to push the rock and make way. However, once the player gets the “Duck Stone”, some rocks may activate once the player is close enough, transforming into a basic enemy Lynx which charges the player. This makes for a more cautious game, where the player must be wary of his environment and not wander too close to any rocks he encounters.

Locks & Keys:

These separate actor's only use is to create an obstacle for the player and force him to search/find/gain a key that opens the correct path-blocking lock. They are also physical representations of a closed door, once the correct key is picked up the locks disappear and clear the way.

Healing

There is no direct way of healing, however at the very beginning of the game the Duck King gives our player character a blessing, which heals him for 1 health point for every 3 defeated enemies.

Levels:

Overview

I wanted to create something special in GB studio, and with the Duck Hero my goal was to create a game similar to the very early Legend of Zelda, where the player can wander off in a large “Open world” kind of level and defeat different enemies. However, GB studio only authorize 20 actors at a time per level, and this also account for pinned actors (Health bar, HUD, Arrow HUD...) and visual only actors (The lock and Key, which don't actually act on their own but is used more like a physical representation of a closed door).

My solution for this was to create huge levels separated into different “rooms” which fits the size of the GameBoy's screen. When the player reaches the end of the screen, he is teleported to the next screen and all actors within that screen is also teleported, placed, and initialized on the next screen, creating the illusion of a vast world with hundreds of enemies, whereas there is actually a maximum of 5 enemy per level at all times.

Level1: The Duck Castle

This level serves as a tutorial for the player, and explains how to use the Sword, the Bow, the Shield and the Hammer. The player also meets the king which sends him on his journey. There is definitely very few intense moments in this level, and the amount of dialogue and hand-holding might slow down the pace of the game, but it's purpose is to gently embark on the player on his journey before the action gets more intense on later levels.

Level2: The Countryside

Here is where things really start. There is no more hand-holding in this level and the player is finally released in a big level where he may go wherever he wants freely. This is also the first level where the player encounters a Lock and the Bunker mini game which is explained further in a later part.

Level3: The Enemy border

These levels feature even more enemies and is possibly the longest level of the game, sending the player left and right to get multiple keys or access for multiple closed doors.

Level4: The Forteress

This final level has a little bit less freedom than the others, but it uses it to create tension: the player starts the level right in front of his ultimate goal: the Lynx Emperor's room, and he needs to get 2 keys to open it. While fetching the keys the tension rises until the final confrontation to save the Duck Kingdom !

MiniGames:

Overview

Originally it was planned that there would be multiple different mini games: there are 4 different weapons and 4 different levels, and so it was originally designed that each level would have its own mini game based on the mechanic of one of the Duck Knight's weapons. However, this proved to be too time-consuming and out of scope, so in the end I opted for a single mini game based on the bow.

The Bunker

This mini game pins the player in the middle of a room and sends waves of enemies toward him. To defend himself, the player needs to shoot down the enemies with his bow before they reach him. After a set number of time/shots the level ends and the player is rewarded for making it out alive. This mini game is played 3 times: once in Level2 (the Countryside), once in Level3 (The Enemy Border) where the player has to survive around 10 seconds, and finally once in the final Level (The Fortress) where the player must shoot the Lynx Emperor 30 times before he is defeated.

Art:

As I am a programmer and not an artist, the art of the game was kept relatively simple. There was also the problematic of limited sprites, and since I wanted to make a “Zelda” like game with huge levels, I also wanted a lot of actors, however Actors and Level unique tile counts toward the same limit, so the total unique tile count is reached quite fast. Because of this limitation I had to limit my art, and especially my animation, substantially. Most levels only contain bare walls and empty plain ground, because intricate walls and grass were too heavy for GB studio.

Most art is made in a cell shaded manner, meaning that all parts of the environment and actors are coated in a black highlight to differentiate the actor from the background.

The Ducks and Lynx are my original creation, but for the background tiles like rocks, bricks and trees I took inspiration from the assets found in the default GB studio game

Music:

The music of the game was made mostly by Victor Vicente.

The goal for the music was to make it sound light and adventurous, while keeping the “knight” theme by making it medieval.

There is a unique music for the Main Menu, levels 1,2 and 4, as well as a short jingle for the BTS logo screen when the game boots up, and for the Game Over screen when the player loses.

Due to time constraints, Level3 has the same music as the Main Menu.